



Taeniasis — *Taenia* spp.

Microbiology GIT Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

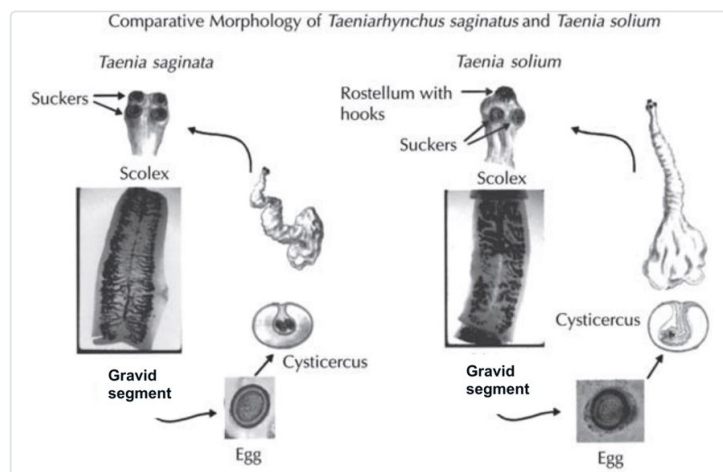
1. Which *Taenia* species causes cysticercosis, and is it acquired by eating undercooked meat or by swallowing eggs?
2. For *T. saginata* and *T. solium*: who is the definitive host, and who is the intermediate host?
3. What is the most common symptom of neurocysticercosis, and which imaging tests detect it best?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 The two tapeworms — one discriminator grid

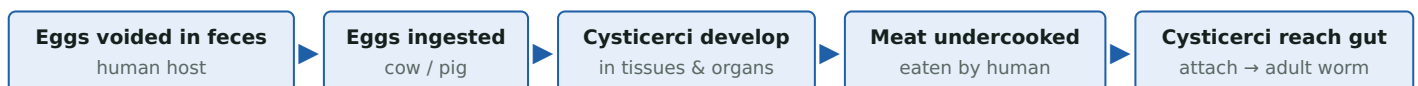
	<i>Taenia saginata</i>	<i>Taenia solium</i>
Common name	Beef tapeworm	Pork tapeworm
Definitive host	Human	Human
Intermediate host	Cow	Pig
Larval (infective) stage	<i>Cysticercus bovis</i>	<i>Cysticercus cellulosae</i>
Scolex (lecture figure)	4 suckers only	4 suckers + rostellum with hooks
Disease in humans	Taeniasis (adult worm in gut)	Taeniasis + cysticercosis (larvae in tissues) — can be fatal

Only 2 of ~45 *Taenia* species commonly infect humans (*T. asiatica* is also named). The **ONE discriminator** the lecture stresses: cysticercosis = *T. solium*.

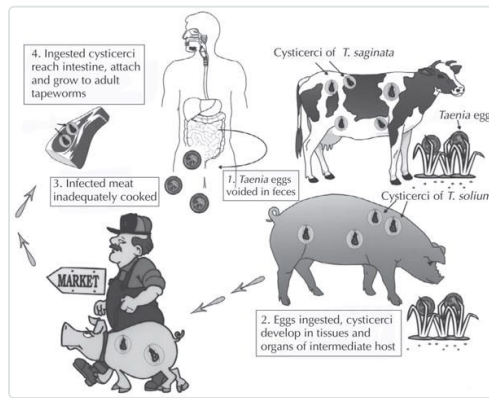


Comparative morphology — scolex, gravid segment, egg and cysticercus of each species.

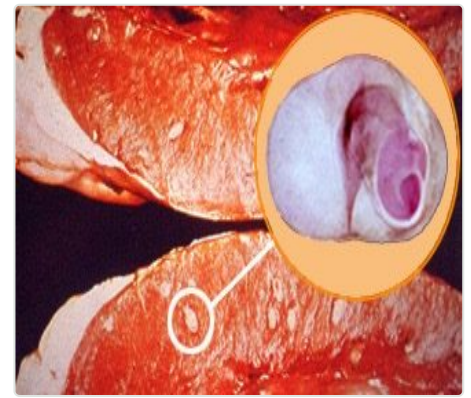
2 Life cycle — how each disease is acquired



Taeniasis = eating the cysticerci in meat. Cysticercosis = a human swallowing *T. solium* eggs (accidental ingestion or autoinfection) so the human becomes the intermediate host.



Life cycle of *T. saginata* (cow) and *T. solium* (pig).



Cysticercus in muscle (circled).

3 Taeniasis vs cysticercosis

	Taeniasis	Cysticercosis
What it is	Infection with the adult worm	Tissue infection by the larval stage (metacestode) of <i>T. solium</i>
How acquired	Undercooked infected beef / pork	Accidental ingestion of eggs or autoinfection
Incubation	6–8 weeks	—
Picture	Minor GI irritation: nausea, vomiting, diarrhea, constipation, hunger pain, chronic indigestion; proglottids passed in feces	Depends on site and number of cysticerci; most often skeletal muscle, eye, brain

4 Cysticercosis by site

Site	What you see
Skeletal muscle / subcutaneous	Usually asymptomatic ; cysts 10–15 mm; destruction of a cyst → strong inflammation, local pain and swelling
Eye	Eye pain, blurred vision , partial or complete loss of vision
Brain — neurocysticercosis (NCC)	Most common parasitic infection of the CNS (60–90% of infected individuals); cysts 4–20 mm; dying parasite → inflammation, granuloma, fibrosis → sub-acute encephalitis. Seizure is the most common symptom ; also headache, dizziness, involuntary movements, raised intracranial pressure, dementia; some never develop symptoms

NCC type	Location	Outcome
Parenchymal	Cysts in brain tissue	Mortality <2% (viable or calcified, single / low burden); a single lesion has an excellent chance of full recovery
Extra-parenchymal / subarachnoid	Spaces around the brain or ventricles	Mortality ~16.6% to >20% if complicated by severe inflammation, raised pressure or stroke

Epidemiology hook: *T. solium* causes ~30% of epilepsy cases in endemic areas where people and roaming pigs live in close proximity.

5 How the cyst survives — immune evasion

Cyst product	What it does
Taeniastatin	Serine-threonine protease inhibitor: blocks complement activation , blocks cytokine production, interferes with neutrophil function
Paramyosin	Impairs complement mainly by inhibiting C1q
Sulfated polysaccharide	Directs activated complement away from the parasite
Host antibodies	Bind the metacestode through Fc receptors, then are degraded and used as an amino-acid source
Glutathione-S-transferase + small molecules	Detoxify reactive oxygen and nitrogen intermediates

Net effect: living cysts provoke little or no inflammation — trouble starts when the parasite is destroyed.

6 Diagnosis · treatment · prevention

	Taeniasis	Cysticercosis
Diagnosis	Proglottids or eggs in a fecal specimen	Serology + epidemiological data; MRI and CT are the most sensitive for NCC
Treatment	Praziquantel 5–10 mg/kg single oral dose; or niclosamide 2 g (adult) in divided doses over 2 days	Praziquantel, albendazole, flubendazole + anti-inflammatory + anticonvulsants
Prevention	Cook beef and pork thoroughly (above 56 °C) or freeze at –10 °C for 10 days	

P Must-remember pearls

- Human = **definitive** host of both species; **cow** (*T. saginata*, *C. bovis*) and **pig** (*T. solium*, *C. cellulosae*) are intermediate hosts.
- **Undercooked meat** → taeniasis (adult worm). **Swallowed *T. solium* eggs** (or autoinfection) → cysticercosis.
- **Only *T. solium*** causes cysticercosis; its scolex carries a **rostellum with hooks**.
- NCC: **seizure** is the commonest symptom; **MRI/CT** detect it best; **extra-parenchymal** disease is far deadlier than parenchymal.
- Praziquantel for the gut worm; antiparasitic + **anti-inflammatory** + **anticonvulsant** for cysticercosis.

Q Test yourself

- 1. A 30-year-old farmer from a village where pigs roam freely presents with new-onset seizures; CT shows cystic lesions in brain tissue. Which route of infection best explains this?**
A. Eating undercooked beef containing *C. bovis*
B. Eating undercooked pork containing *C. cellulosae*
C. Accidental ingestion of *T. solium* eggs
D. Skin penetration by larvae from soil
- 2. After treatment, a scolex is recovered from a patient who passed white segments in stool. It shows four suckers and a rostellum with hooks. What is the species and its intermediate host?**
A. *T. saginata*; cow
B. *T. saginata*; pig
C. *T. solium*; cow
D. *T. solium*; pig
- 3. Which finding best separates *T. solium* from *T. saginata* in terms of human disease?**
A. Only *T. solium* infects humans as adult worms
B. Only *T. solium* larvae cause cysticercosis
C. Only *T. saginata* passes proglottids in feces
D. Only *T. saginata* is acquired from meat
- 4. Which drug is given as a single oral dose of 5–10 mg/kg for adult intestinal *Taenia* infection?**
A. Praziquantel
B. Pyrantel pamoate
C. Levamisole
D. Thiabendazole
- 5. Which neurocysticercosis pattern carries the worst prognosis?**
A. A single viable cyst in brain tissue
B. Calcified parenchymal lesions
C. Extra-parenchymal (subarachnoid / ventricular) cysts
D. Asymptomatic muscle cysts

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 — C** — Cysticercosis follows swallowing eggs (or autoinfection); eating meat cysticerci causes taeniasis, not cysticercosis.
- 2 — D** — A rostellum with hooks marks *T. solium*, the pork tapeworm (pig is the intermediate host).
- 3 — B** — Both cause taeniasis; cysticercosis is caused by the larval stage of *T. solium*.
- 4 — A** — Praziquantel (5–10 mg/kg once) is the lecture's first drug for taeniasis.
- 5 — C** — Extra-parenchymal disease has a reported mortality of ~16.6% to >20% when complicated; parenchymal disease is <2%.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam